

MACHINE LEARNING COURSE · CS-433

Generalization, Model Selection, and Validation

September 29, 2026

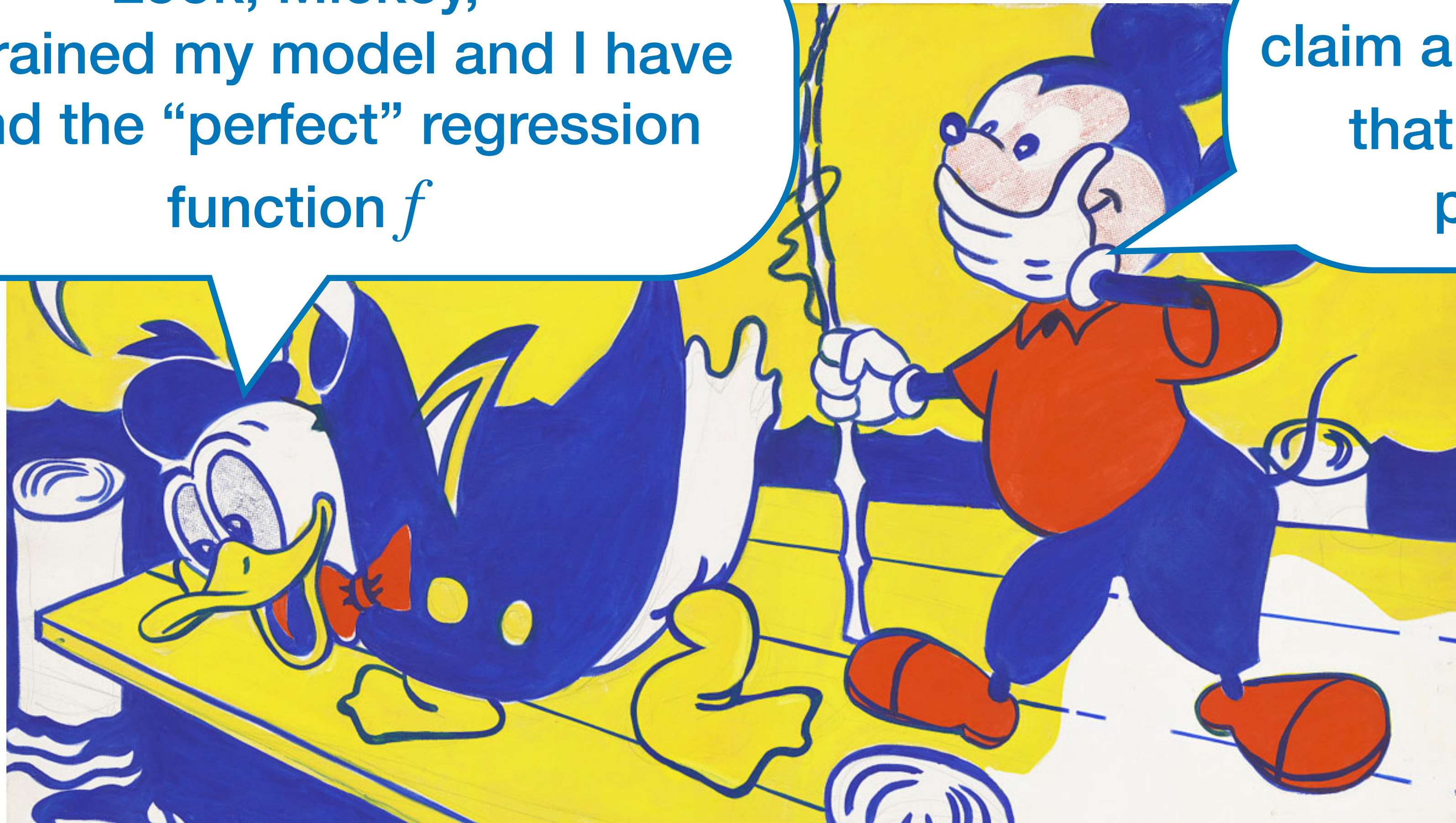
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Generalization, validation?

Look, Mickey,
I've trained my model and I have
found the “perfect” regression
function f

How can I verify this
claim and have confidence
that f will have good
performance?



What is the model selection problem?

- Ridge regression: $w_\lambda = \arg \min_w \frac{1}{2N} \sum_{n=1}^N (y_n - x_n^\top w)^2 + \lambda \|w\|_2^2$
Hyperparameter
 - λ can be tuned to control the model complexity (to reduce overfitting)
 - In practice: $(\lambda_1, \dots, \lambda_k) \longrightarrow \text{Algorithm} \longrightarrow (w_1, \dots, w_k)$
 - Which λ should we use?
- Polynomial feature expansion: $(x_{(1)}, x_{(2)}) \xrightarrow{\phi} (x_{(1)}, x_{(2)}, x_{(1)}^2 + x_{(2)}^2)$
 $(x_{(1)}, x_{(2)}, 5x_{(1)}^2 + 2x_{(2)}^2, x_{(2)}^3 + 2x_{(1)})$
 - Enrich the model complexity, by augmenting the feature vector x .
 - Here the degree d is the hyperparameter

We are facing the same problem: **how do we choose these hyperparameters?**

Model selection for neural networks

Algorithms?

SGD
Adam
Which step-size?
Which batch-size?
Which momentum?

Architectures?

FullyConnected
ConvNet
ResNet
Transformer
Which width?
Which depth?
Batch normalization?

Regularizations?

Weight decay?
Early stopping?
Data augmentations?

Probabilistic Setup

Data Model:

Unknown distribution $\mathcal{D}$ with range $\mathcal{X} \times \mathcal{Y}$

We see a dataset S of independent samples from $\mathcal{D}$:

$$S = \{(x_n, y_n)\}_{n=1}^N \sim \mathcal{D} \quad \text{i.i.d.}$$

Learning Algorithm:

$$\begin{array}{ccc} & \mathcal{A}(S) = f_S & \\ \nearrow \text{Input} & & \nwarrow \text{Output} \end{array}$$

Ridge regression: gradient descent or least-squares estimator

Can add a subscript $f_{S,\lambda}$ to indicate the hyper parameter dependency

Generalization Error: how accurate is f at predicting?

We compute the **expected error** over all samples drawn from distribution $\mathcal{D}$:

$$L_{\mathcal{D}}(f) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(y, f(x))]$$

where $\ell(\cdot, \cdot)$ is the loss function

- Ex: $\ell(y, y') = \frac{1}{2}(y - y')^2$, logistic loss, hinge loss

The quantity $L_{\mathcal{D}}(f)$ has many names: $\left\{ \begin{array}{l} \text{True} \\ \text{Expected} \\ \text{Generalization} \end{array} \right\} \left\{ \begin{array}{l} \text{Risk} \\ \text{Error} \\ \text{Loss} \end{array} \right\}$

This is the quantity we are fundamentally interested in

 Problem: $\mathcal{D}$ is unknown

Empirical Error: what we can compute

We can approximate the true error by **averaging the loss function over the dataset**

$$L_S(f) = \frac{1}{|S|} \sum_{(x_n, y_n) \in S} \ell(y_n, f(x_n)) .$$

Also called: **empirical risk/error/loss**

 The samples are random thus $L_S(f)$ is a random variable

It is an unbiased estimator of the true error

Law of large number: $L_S(f) \xrightarrow{|S| \rightarrow \infty} L_{\mathcal{D}}(f)$ but fluctuations!

Generalization gap: $|L_{\mathcal{D}}(f) - L_S(f)|$

Training error: what we are minimizing

△ the prediction function f_S is itself a function of the data S

When the model has been trained on the same data it is applied to, the empirical error is called the ***training error***:

$$L_S(f_S) = \frac{1}{|S|} \sum_{(x_n, y_n) \in S} \ell(y_n, f_S(x_n))$$

This is the objective function you are minimizing to find the predictor

It might not be representative of the error we see on “fresh” samples

The reason that $L_S(f_S)$ might not be close to $L_{\mathcal{D}}(f_S)$ is overfitting

Splitting the data

Problem: Validating model on the same data we trained it on

Fix: **Split the data** into an independent *training and test set*:

$$S = S_{\text{train}} \cup S_{\text{test}}$$

1. We **learn** the function $f_{S_{\text{train}}}$ using the **train set**
2. We **validate** it computing the error on the **test set**

$$L_{S_{\text{test}}}(f_{S_{\text{train}}}) = \frac{1}{|S_{\text{test}}|} \sum_{(y_n, x_n) \in S_{\text{test}}} \ell(y_n, f_{S_{\text{train}}}(x_n))$$

➡ Since S_{test} and S_{train} are independent: $L_{S_{\text{test}}}(f_{S_{\text{train}}}) \approx L_{\mathcal{D}}(f_{S_{\text{train}}})$

 We have less data both for the learning and the validation tasks (tradeoff)

Generalization gap: How far is the test from the true error?

Claim: given a model f and a test set $S_{\text{test}} \sim \mathcal{D}$ i.i.d. (not used to learn f) and a loss $\ell(\cdot, \cdot) \in [a, b]$:

$$\mathbb{P} \left[\underbrace{\left| L_{\mathcal{D}}(f) - L_{S_{\text{test}}}(f) \right|}_{\text{Generalization Gap}} \geq \sqrt{\frac{(b-a)^2 \ln(2/\delta)}{2 |S_{\text{test}}|}} \right] \leq \delta$$

The error decreases as $\mathcal{O}(1/\sqrt{|S_{\text{test}}|})$ with the number of test points

High probability bound: δ is only in the $\ln$

➡ The more data points we have, the more confident we are that the empirical loss we measure is close to the true loss

Why do you care?

- Given a predictor f and a dataset S you can control the expected risk:

$$\mathbb{P}\left(\underbrace{L_{\mathcal{D}}(f)}_{\text{not computable}} \geq \underbrace{L_{S_{\text{test}}}(f)}_{\text{Computable}} + \underbrace{\sqrt{\frac{(a-b)^2 \ln(2/\delta)}{2|S_{\text{test}}|}}}_{\text{deviation}}\right) \leq \delta$$

- Given a dataset S
 - Split: $S = S_{\text{train}} \cup S_{\text{test}}$
 - Train: $\mathcal{A}(S_{\text{train}}) = f_{S_{\text{train}}}$
 - Use:

$$\mathbb{P}\left(L_{\mathcal{D}}(f_{S_{\text{train}}}) \geq L_{S_{\text{test}}}(f_{S_{\text{train}}}) + \sqrt{\frac{(a-b)^2 \ln(2/\delta)}{2|S_{\text{test}}|}}\right) \leq \delta$$

➡ We can obtain a probabilistic upper bound on the expected risk

The proof relies only on concentration inequalities

Since $(x_n, y_n) \in S_{\text{test}}$ are chosen independently, the associated losses $\Theta_n = \ell(y_n, f(x_n)) \in [a, b]$ given a fixed model f , are also i.i.d. random variables

Empirical loss:
$$\frac{1}{N} \sum_{n=1}^N \Theta_n = \frac{1}{N} \sum_{n=1}^N \ell(y_n, f(x_n)) = L_{S_{\text{test}}}(f)$$

True loss:
$$\mathbb{E}[\Theta_n] = \mathbb{E}[\ell(y_n, f(x_n))] = L_{\mathcal{D}}(f)$$

What is the chance that the empirical loss $L_{S_{\text{test}}}(f)$ deviates from the true loss by more than a given constant?

➔ classically addressed using **concentration inequalities**

Hoeffding inequality: a simple concentration bound

Claim: Let $\Theta_1, \dots, \Theta_N$ be a sequence of i.i.d. random variables with mean $\mathbb{E}[\Theta]$ and range $[a, b]$

$$\mathbb{P} \left[\left| \frac{1}{N} \sum_{n=1}^N \Theta_n - \mathbb{E}[\Theta] \right| \geq \varepsilon \right] \leq 2e^{-2N\varepsilon^2/(b-a)^2} \text{ for any } \varepsilon \geq 0$$

Concentration bound: the empirical mean is concentrated around its mean

A. Use it with $\Theta_n = \ell(y_n, f(x_n))$

B. Equating $\delta = 2e^{-2|S_{\text{test}}|\varepsilon^2/(b-a)^2}$ we get $\varepsilon = \sqrt{\frac{(b-a)^2 \ln(2/\delta)}{2|S_{\text{test}}|}}$ $\square$

Model Selection: pick the best model

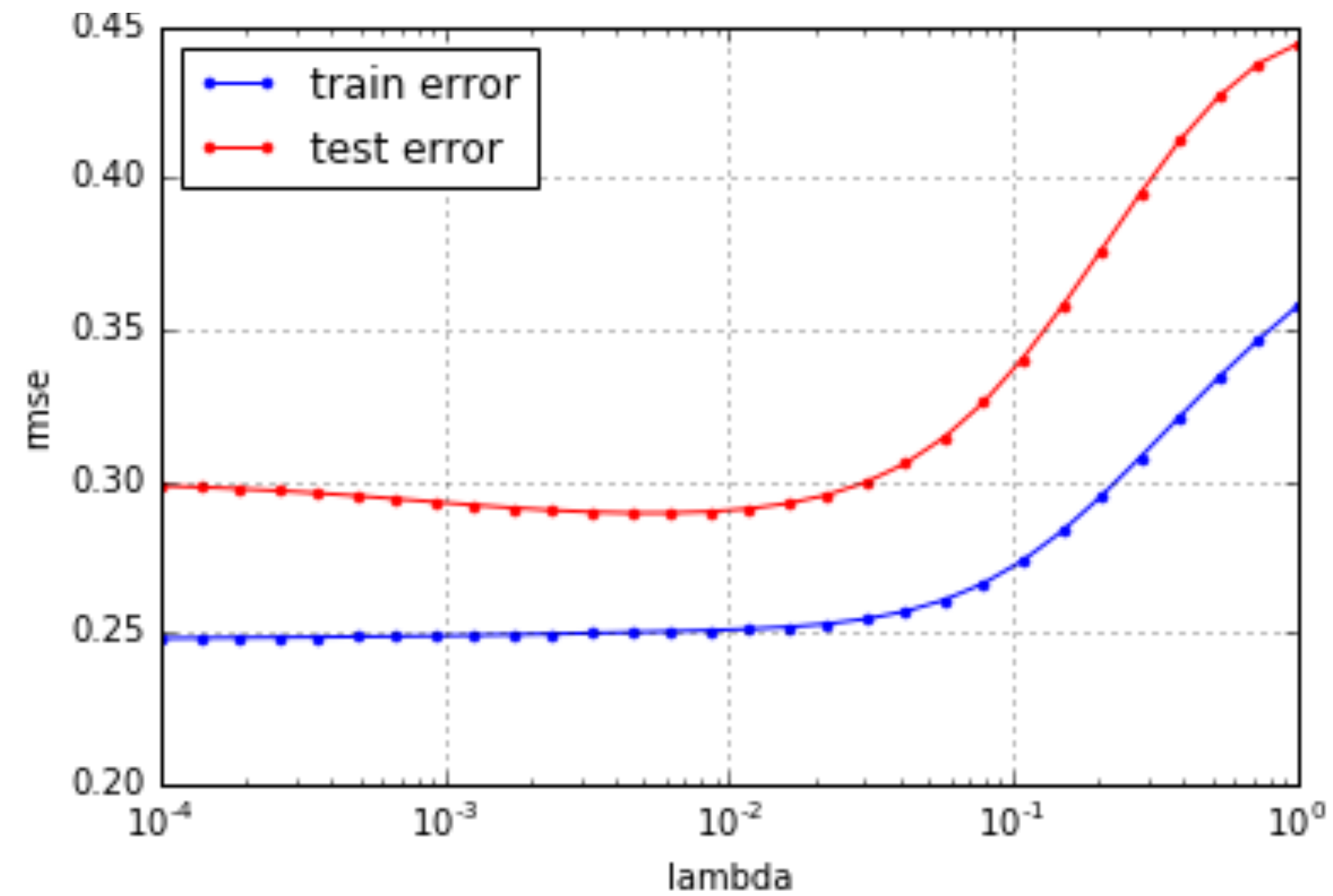
Goal: select the hyperparameters of our model (λ for ex. in ridge regression)

We have a set of candidate values $\{\lambda_k\}_{k=1}^K$. Which one should we choose?

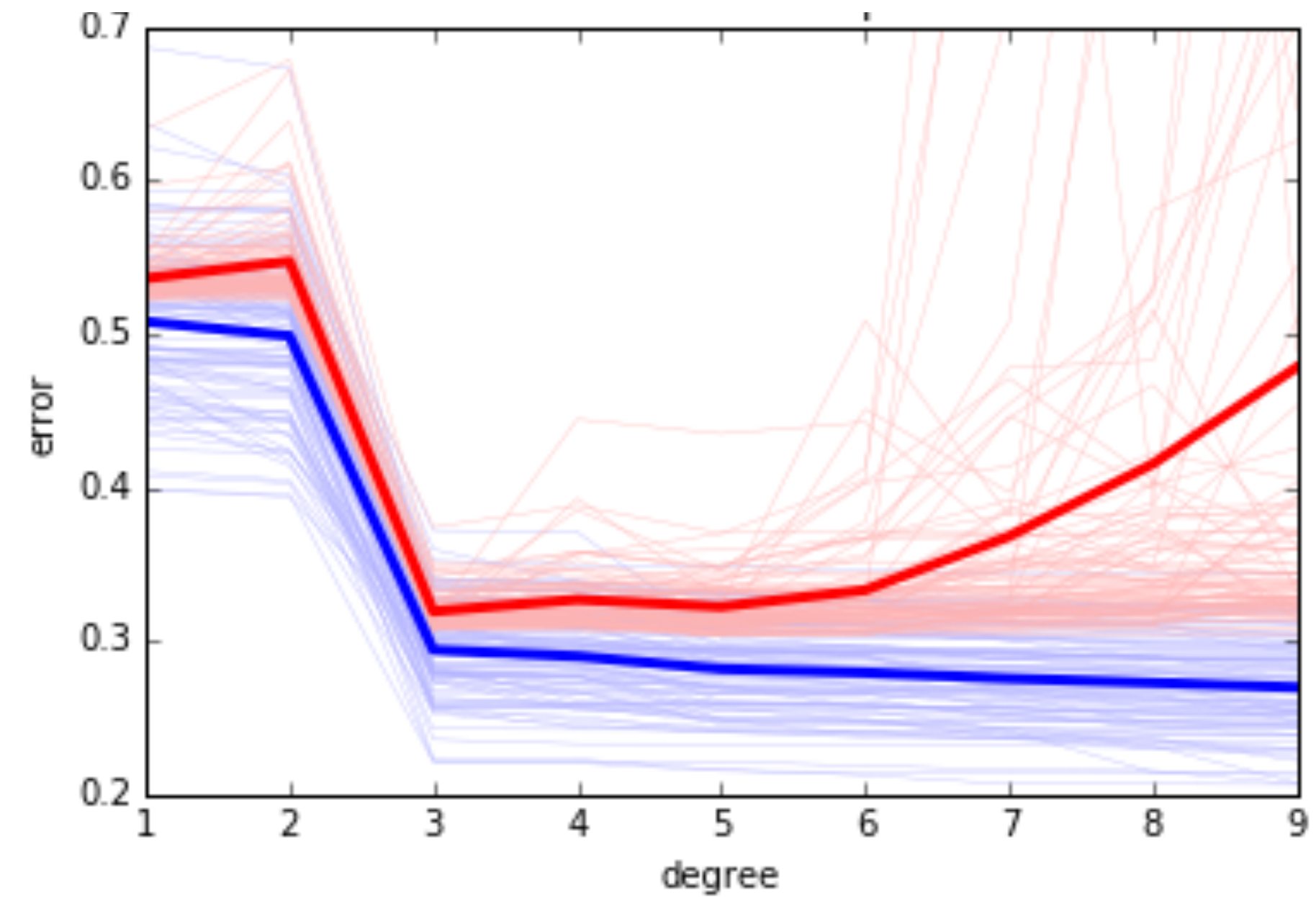
1. Split the data into $S = S_{\text{train}} \cup S_{\text{test}}$, generated independently from $\mathcal{D}$
2. Run the learning algorithm K times on the same training set S_{train} to compute the K prediction functions $f_{S_{\text{train}}, \lambda_k}$
3. For each prediction function, compute the test error $L_{S_{\text{test}}}(f_{S_{\text{train}}, \lambda_k})$

We then choose the value of the parameter λ giving the smallest test error

Examples



Ridge regression



Degree in case of a polynomial feature expansion

Does model selection work?

Two questions:

- How do we know that the best function $f_{S_{\text{train}}, \lambda}$ is a good approximation of the best model within our function class?
- How do we know that $L_{S_{\text{test}}}(f_{S_{\text{train}}, \lambda_k}) \approx L_{\mathcal{D}}(f_{S_{\text{train}}, \lambda_k})$?
 - We have discussed it for a single model
 - What about several models?
 - I.e., what is the justification that the min is actually good?

How far is each of the K test errors $L_{S_{\text{test}}}(f_k)$ from the true $L_{\mathcal{D}}(f_k)$?

Claim: we can bound the maximum deviation for all K candidates, by

$$\mathbb{P} \left[\max_k \left| L_{\mathcal{D}}(f_k) - L_{S_{\text{test}}}(f_k) \right| \geq \sqrt{\frac{(b-a)^2 \ln(2K/\delta)}{2|S_{\text{test}}|}} \right] \leq \delta$$

- The error decreases as $\mathcal{O}(1/\sqrt{|S_{\text{test}}|})$ with the number test points
 - When testing K hyper-parameters, the error only goes up by $\sqrt{\ln(K)}$
- ➡ So we can test many different models without incurring a large penalty
- It can be extended to infinitely many models

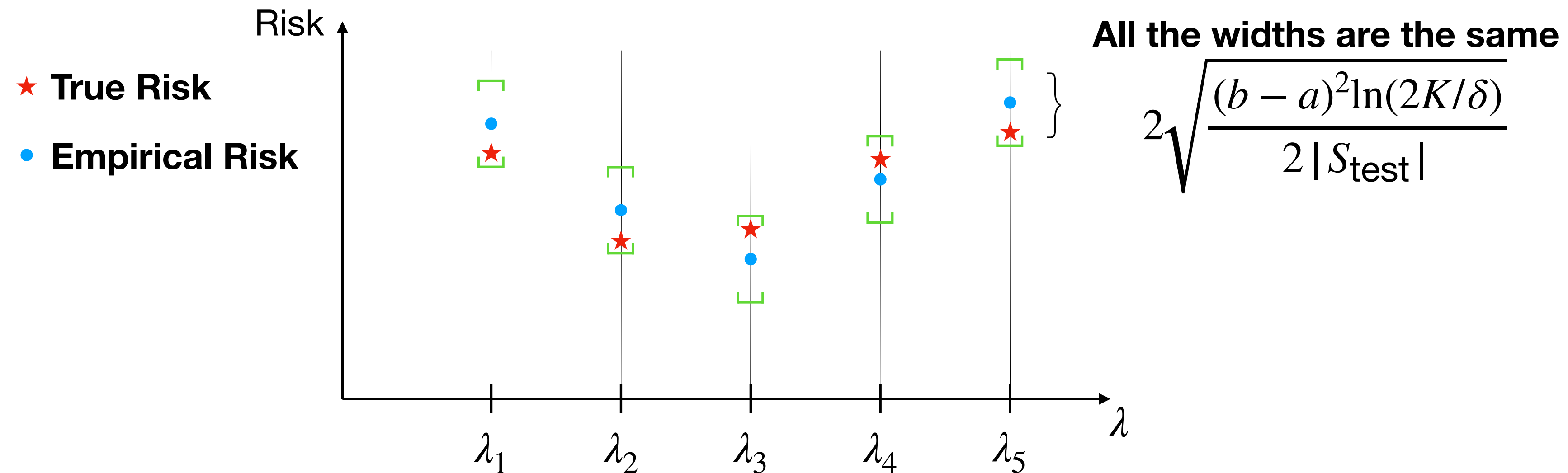
Proof: A simple union bound

The proof of this statement follows the proof of the special case $K = 1$

$$\begin{aligned}\mathbb{P}\left[\max_k \left|L_{\mathcal{D}}(f_k) - L_{S_{\text{test}}}(f_k)\right| \geq \varepsilon\right] &= \mathbb{P}\left[\cup_k \left\{\left|L_{\mathcal{D}}(f_k) - L_{S_{\text{test}}}(f_k)\right| \geq \varepsilon\right\}\right] \\ &\leq \sum_k \mathbb{P}\left[\left|L_{\mathcal{D}}(f_k) - L_{S_{\text{test}}}(f_k)\right| \geq \varepsilon\right] \\ &\leq 2Ke^{-2N\varepsilon^2/(b-a)^2}\end{aligned}$$

Hence, equating $\delta = 2Ke^{-2N\varepsilon^2/(b-a)^2}$, we get $\varepsilon = \sqrt{\frac{(b-a)^2 \ln(2K/\delta)}{2N}}$ as stated

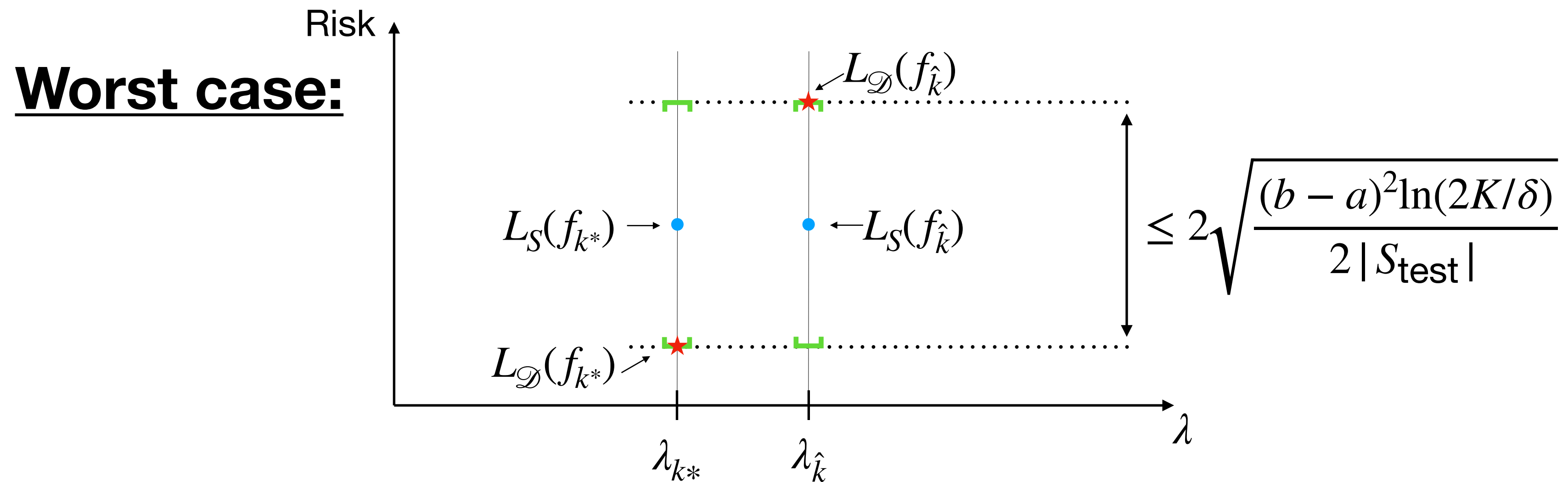
If we choose the “best” function according to the empirical risk then its true risk is not too far away from the true risk of the optimal choice



Let $k^* = \operatorname{argmin}_k L_{\mathcal{D}}(f_k)$ and $\hat{k} = \operatorname{argmin}_k L_{S_{\text{test}}}(f_k)$ then

$$\mathbb{P} \left[\underbrace{L_{\mathcal{D}}(f_{\hat{k}})}_{\text{Function with the smallest empirical risk}} \geq \underbrace{L_{\mathcal{D}}(f_{k^*})}_{\text{Function with the smallest true risk}} + 2\sqrt{\frac{(b-a)^2 \ln(2K/\delta)}{2|S_{\text{test}}|}} \right] \leq \delta$$

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Function with
the smallest empirical risk

Function with
the smallest true risk

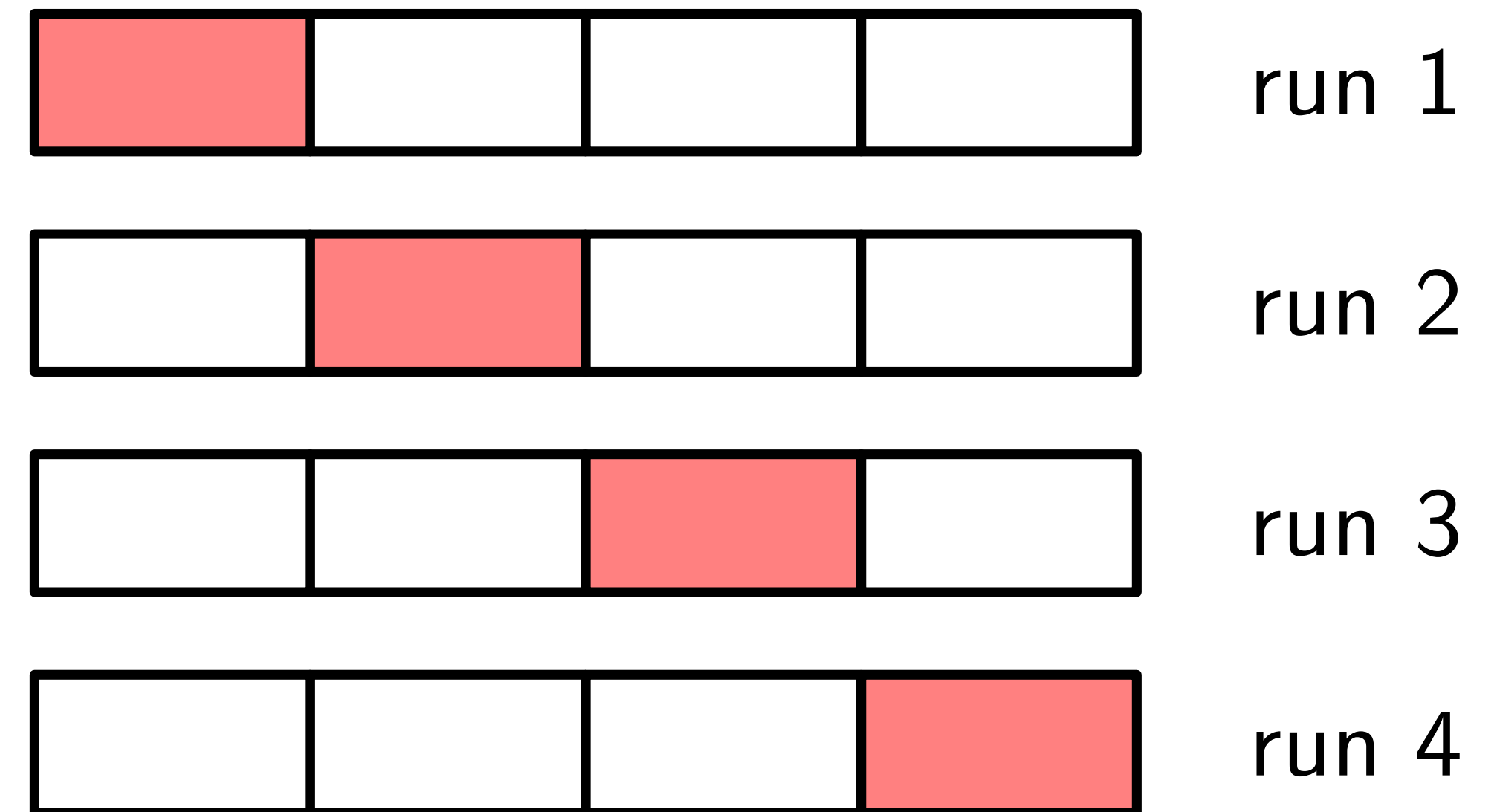
Cross-Validation



- Splitting the data once into two parts (one for training and one for testing) is not the most efficient way to use the data
- Cross-validation is a better way

M-fold Cross-Validation*

1. Randomly partition the data into M groups
2. Train M times. Each time leave out exactly one of the M groups for testing and use the remaining $M - 1$ groups for training.
3. Average the M results



- We have used all data for training, and all data for testing, and used each data point the same number of times
 - Cross-validation returns an estimate of the *generalization-error* and its **variance**
- * Usually called “K-fold”, but that would clash with K as used before (# values tried for hyperparam)...

Do we still have some time?

Hoeffding's inequality:

Let $\Theta_1, \dots, \Theta_N$ be a sequence of i.i.d. random variables with mean $\mathbb{E}[\Theta]$ and range $[a, b]$. Then, for any $\varepsilon > 0$

$$\mathbb{P} \left[\left| \frac{1}{N} \sum_{n=1}^N \Theta_n - \mathbb{E}[\Theta] \right| \geq \varepsilon \right] \leq 2e^{-2N\varepsilon^2/(b-a)^2}$$

Proof (I)

- We equivalently assume that $\mathbb{E}[\Theta] = 0$ and that $\Theta_n \in [a, b]$
- We will only show that

$$\mathbb{P}\left\{\frac{1}{N}\sum_{n=1}^N \Theta_n \geq \varepsilon\right\} \leq e^{-2N\varepsilon^2/(b-a)^2}$$

This, together with the equivalent bound

$$\mathbb{P}\left\{\frac{1}{N}\sum_{n=1}^N \Theta_n \leq -\varepsilon\right\} \leq e^{-2N\varepsilon^2/(b-a)^2}$$

will prove the claim

Proof (II)

For any $s \geq 0$,

$$\begin{aligned}\mathbb{P}\left\{\frac{1}{N}\sum_{n=1}^N\Theta_n \geq \varepsilon\right\} &= \mathbb{P}\left\{s\frac{1}{N}\sum_{n=1}^N\Theta_n \geq s\varepsilon\right\} \\ &= \mathbb{P}\left\{e^{s\frac{1}{N}\sum_{n=1}^N\Theta_n} \geq e^{s\varepsilon}\right\} \\ &\leq \mathbb{E}[e^{s\frac{1}{N}\sum_{n=1}^N\Theta_n}]e^{-s\varepsilon} && \text{(Markov inequality)} \\ &= \prod_{n=1}^N \mathbb{E}[e^{\frac{s\Theta_n}{N}}]e^{-s\varepsilon} && \text{(the r.v. } \Theta_n \text{ are independent)} \\ &= \mathbb{E}[e^{\frac{s\Theta}{N}}]^N e^{-s\varepsilon} && \text{(the r.v. } \Theta_n \text{ are i.d.)} \\ &\leq e^{s^2(b-a)^2/(8N)} e^{-s\varepsilon} && \text{(Hoeffding lemma)}\end{aligned}$$

Proof (III)

What do we do now? We have for any $s \geq 0$

$$\mathbb{P}\left\{\frac{1}{N}\sum_{n=1}^N\Theta_n \geq \varepsilon\right\} \leq e^{s^2(b-a)^2/(8N)}e^{-s\varepsilon}$$

In particular for the minimum value obtained for $s = \frac{4N\varepsilon}{(b-a)^2}$

$$\mathbb{P}\left\{\frac{1}{N}\sum_{n=1}^N\Theta_n \geq \varepsilon\right\} \leq e^{-2N\varepsilon^2/(b-a)^2}$$

Hoeffding lemma

For any random variable X , with $\mathbb{E}[X] = 0$ and $X \in [a, b]$ we have

$$\mathbb{E}[e^{sX}] \leq e^{\frac{1}{8}s^2(b-a)^2} \text{ for any } s \geq 0$$

Proof outline:

Consider the convex function $s \mapsto e^{sx}$. In the range $[a, b]$ it is upper bounded by the chord

$$e^{sx} \leq \frac{x-a}{b-a}e^{sb} + \frac{b-x}{b-a}e^{sa}$$

Taking the expectation and recalling that $\mathbb{E}[X] = 0$, we get

$$\mathbb{E}[e^{sX}] \leq \frac{b}{b-a}e^{sa} - \frac{a}{b-a}e^{sb} \leq e^{s^2(b-a)^2/8}$$